# Experiment 4

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## Title

Develop a dynamic webpage where content updates on button click using JavaScript DOM. Implement jQuery effects: hide/show elements, toggle visibility, and animate elements.

## Theory:

1. **JavaScript and the DOM (Document Object Model):**
   - The **DOM** is a hierarchical representation of the elements on a webpage. It allows programming languages like JavaScript to interact with the structure, style, and content of the page.
   - Using **DOM methods**, we can:
     - Access elements (getElementById, getElementsByClassName, querySelector)
     - Change content (innerHTML, textContent)
     - Modify attributes (setAttribute, removeAttribute)
     - Change styles dynamically (style.property)
   - Example: Changing the text of a paragraph when a button is clicked enhances user interactivity.
2. **Dynamic Webpage Content:**
   - A **dynamic webpage** changes content without reloading the page.
   - Dynamic updates improve **user experience** by allowing real-time feedback and interactivity.
   - In this lab, content updates are achieved using document.getElementById().innerHTML to manipulate text dynamically.
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3. **Introduction to jQuery:**
   - **jQuery** is a lightweight, fast JavaScript library that simplifies:
     - HTML document traversal and manipulation

- Event handling
- Animations and effects

- jQuery is **cross-browser compatible**, meaning the same code works in different browsers without modification.

Syntax Example:
$(selector).action();

- $() is the jQuery function
- selector identifies the HTML element
- action() is the method applied (like hide(), show(), animate())

4. **jQuery Effects and Animations:**
   - **Hide/Show:** Instantly hide or show elements.
   - **Toggle:** Switch visibility state of an element (hide → show, show → hide).
   - **Animate:** Allows smooth transitions of CSS properties (like height, width, opacity, margins).
   - **Benefits of jQuery Effects:**
     - Improves visual interactivity
     - Reduces manual CSS manipulation
     - Provides smoother user experience

5. **Integration of JS DOM and jQuery:**
   - Using **pure JavaScript** allows control over element content and structure.
   - Using **jQuery** complements JS by handling animations and effects more efficiently.
   - Combining both allows for **dynamic and interactive web pages** where content changes and user interactions are smooth and visually appealing.

6. **Applications:**
   - Dynamic webpages are used in:
     - Online forms that validate inputs instantly
     - Content sliders and carousels
     - Real-time notifications
     - Interactive dashboards

## Procedure

1. Include jQuery CDN:

```
<script src="https://code.jquery.com/jquery-3.7.0.min.js"></script>
```

2. Create HTML element:

```
<div id="box">Hello</div>
<button id="btn">Toggle</button>
```

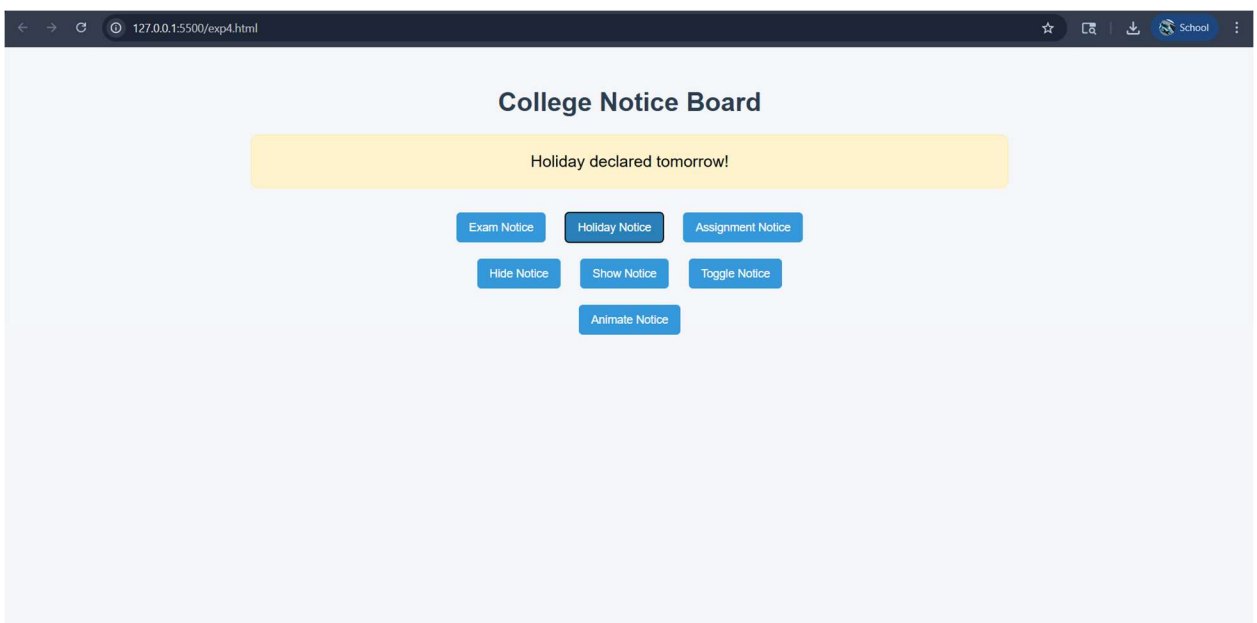
3. Apply jQuery:

```
<script>
$('#btn').click(function(){
  $('#box').toggle(500);
});
</script>
```

**Scenario-Based Exercises:( jQuery Lab Task)**

Design a simple interactive College Notice Board webpage using HTML, CSS, JavaScript, and jQuery. The page should display a default notice message and include buttons that update the notice text dynamically using JavaScript DOM when clicked. Use jQuery to hide and show the notice, toggle its visibility with a single button, and animate the notice by moving it or changing its appearance (such as font size or opacity). The webpage should work without page refresh and demonstrate basic DOM manipulation and jQuery effects.

## Output:



## Conclusion

JavaScript DOM allows dynamic updating of webpage content in real-time. jQuery simplifies element manipulation, effects, and animations. Combining both creates interactive and visually appealing webpages, enhancing user experience. This lab demonstrated practical implementation of content updates, visibility toggling, and animations.

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## Question:

1. You created a PHP file but it displays as plain text in the browser. What is the most likely reason?Ans: **B) File extension is .html instead of .php**

   A) PHP syntax error
   B) File extension is .html instead of .php
   C) MySQL not installed
   D) CSS not linked

2. Write a PHP script to establish a connection with a MySQL database named `college_db`.

```
<?php
$servername = "localhost";
$username = "root";
// default username
$password = "";
// default password (empty in XAMPP)
$dbname = "college_db";

// Create connection
$conn = new mysqli($servername, $username, $password, $dbname);
```

```
// Check connection
if ($conn->connect_error) {
        die("Connection failed: " . $conn->connect_error); }
        echo "Connected successfully!";
?>
```

3. What is the purpose of session_start() in PHP?Ans: **B) Start user session tracking**

A) Connect database
B) Start user session tracking
C) Close server
D) Execute SQL